

Team Name - NER-AI LOGISTICS

RASTA-AI

Roads treated as changeably passable — not as fixed lines on a map.

THE PROBLEM — SPECIFIC TO NER

22 km chokepoint

Siliguri Corridor feeds all 8 NER states.

No parallel road

Next alternative often 100+ km away.

Monsoon closes highways

May–Sep landslides, washouts, flooding.

Dead cellular zones

Trucks unreachable for hours, routinely.

Manager learns by phone call

Needs signal, and a driver able to call.

Fuel budgeted by rule of thumb

Gradient and payload not modelled at all.

PROPOSED SOLUTION

Accessibility intelligence

Weather, landslide and closure evidence revise which corridors are usable, live.

3-corridor explainable compare

Primary / fuel-saving / backup — distance, ETA, litres, reason-coded risk.

Physics fuel model

Litres and CO₂ from payload and climb metres, not distance.

Fleet Sentinel

Stationary + unanswered → automatic SOS briefing to the manager.

Offline-first driver app

Route, turns, POIs and emergency guide cached; GPS buffers and replays.

OUR USP — WHY THIS IS DIFFERENT

“Silence is not evidence.”

No elapsed time can mark a road open. A repair claim is a separate state from a confirmation.

A closed road is refused, not penalised

A score can be out-voted. A refusal cannot.

UNKNOWN ≠ SAFE · STALE ≠ LIVE

Missing data is displayed as missing. No provider outage can become “risk = 0”.

AI is never in the safety path

Rules decide. The model only rewords facts already computed.

Our own trucks are the sensor

A GPS traversal is the only evidence that reopens a road without a bulletin.

DELIVERED AND MEASURED

1,636

automated tests
passing

3

shipped clients
web · Android · API

10

deterministic
decision engines

71 MB

installable
Android APK

17

database tables
in production

ALREADY WORKING IN THE PROTOTYPE

Atomic plan + dispatch

Live GPS freshness

3-route risk compare

Authorised route selection

Human-approved reroute

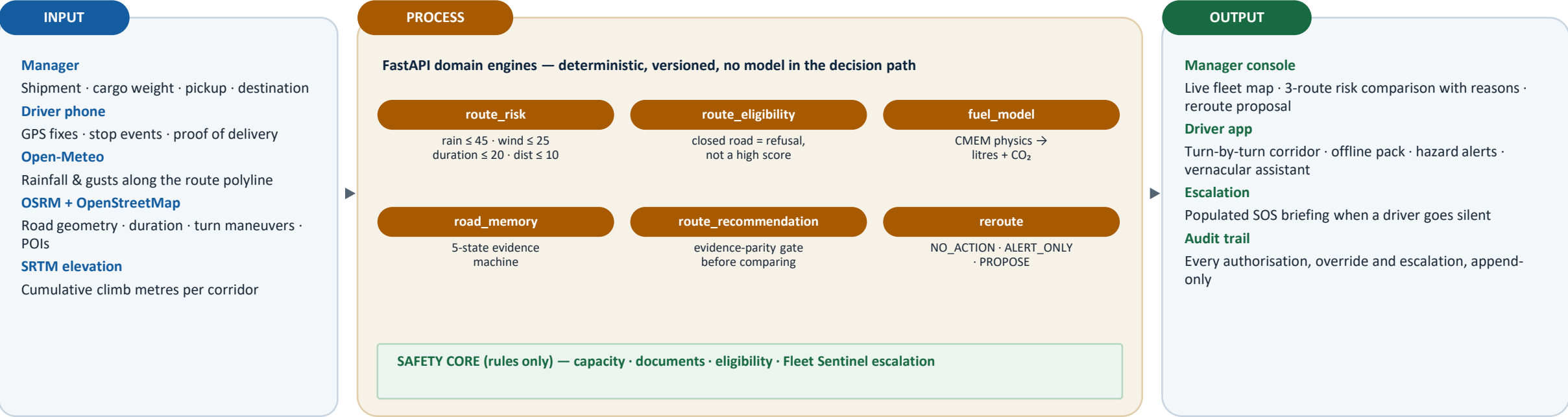
Turn-by-turn nav

Offline package

Vernacular assistant

TECHNICAL APPROACH

How a trip flows through the system — input, process, output



TECHNOLOGIES USED — AND WHY EACH ONE IS REQUIRED						
React + TypeScript Manager console	Expo / React Native One Android build, phone-only field kit	FastAPI · Python 3.11 Async API + the domain engines	PostgreSQL + PostGIS Corridor geometry, ST_DWithin stationary checks	Supabase (ap-south-1) Managed Postgres + edge functions, in-India data	MapLibre + OSM Licence-free fleet and route maps	Gemini Flash (edge) Vernacular restatement only — outside every decision



FEASIBILITY AND VIABILITY

Technology · implementation · cost · scalability



TECHNOLOGY

- Backend, manager console and Android APK all run today.
- 1,636 automated tests pass across three suites.
- No GPU, no on-prem server, no vehicle retrofit.

IMPLEMENTATION

- Driver needs only an Android phone — no telematics box.
- Runs on one corridor pair or a whole state, unchanged.
- Same schema everywhere; only hazard feeds are regional.

COST

- All data sources are free and licence-clear.
- Managed Postgres + a container backend is the whole bill.
- Cost scales with trucks, not with kilometres of road.

SCALABILITY

- Only the hazard feeds are regional; the engine is not.
- Serves Himalayan, Ghat and island logistics unchanged.
- Stateless API; PostGIS indexes the geometry.

CHALLENGES AND RISKS

No source publishes “this road is open again.”

GSI forecasts hazard, ASDMA reports damage, IMD reports rain. Reopening is decided ad hoc by PWD / NHAI / BRO, so no feed carries it.

Connectivity is the operating condition, not the exception.

Any client that assumes a network fails exactly where it matters most.

Free-tier provider outage or quota exhaustion.

A weather failure must never silently become “risk = 0”.

A score alone can be out-voted at the moment it matters most.

Alert fatigue — a gate that fires on everything gets clicked through.

STRATEGIES TO OVERCOME

Our own trucks close the evidence loop.

A GPS traversal reopens a segment when no bulletin can. Every corridor carries an append-only evidence log.

Offline packages and a replay buffer.

Route, turns, POIs and emergency guide cached before the dead zone; GPS queues and replays idempotently.

Fail honest, never fail quiet.

Provider down → factor marked NOT_AVAILABLE and shown in the UI.

Closure is a refusal, not a penalty.

A confirmed closure is rejected at planning time, so no other factor can out-vote it.

Human authorisation for every route change.

Reviewer token required; the driver's map never changes underneath them.

DEPLOYMENT PATH · Guwahati – Shillong – Silchar corridor pair → state-level rollout on the same schema → all eight NER states on one accessibility record.

IMPACT AND BENEFITS

Who benefits, and what changes for them

WHO BENEFITS

FLEET MANAGER

Never has to ask “where is that truck, is the driver okay?” — already answered, or already escalated.

DRIVER

Found when something goes wrong. Route, hazards and emergency help work with no signal, in their language.

OPERATIONS & FINANCE

Trip cost from a physics model, not a rule of thumb. Document and payment state tracked per trip.

MDoNER / STATE TRANSPORT

An auditable record of which NER corridors were passable, and when — today held only as word of mouth.

MEASURABLE IMPACT — WHAT THE PLATFORM DELIVERS

	WITHOUT	WITH RASTA-AI
Detection time for a stranded truck	hours (a phone call)	≤ 90 min, by rule
Routes planned over a closed road	never detected	refused at planning time
Fuel estimate basis	distance only	payload + climb metres
CO ₂ visibility per corridor	none	reported at decision time
Road-condition evidence retained	word of mouth	append-only, per segment

SOCIAL

- Driver silence escalates by rule, not by luck.
- Assamese · Bengali · Hindi · English in-cab guidance.
- 45-minute rest tracking; offline 112 / 108 / 1033.
- Information parity between cab and office.

ECONOMIC

- Fuel planned on payload and climb, not distance.
- Closures refused at planning, not found at the roadblock.
- Over-capacity loading refused before dispatch.
- No per-vehicle hardware cost.

ENVIRONMENTAL

- CO₂ shown per corridor at decision time.
- Avoided detours and hill idling are avoided emissions.
- Load-ratio planning cuts part-loaded running.
- Digital documents and proof of delivery.

DATA SOURCES — LIVE IN THE PLATFORM

Route, terrain and weather evidence

Open-Meteo forecast API — rainfall and gusts · open-meteo.com

Project-OSRM — geometry, duration, turn maneuvers · project-osrm.org

OpenStreetMap / Overpass — roadside POIs · overpass-api.de

NASA SRTM — elevation climb for the fuel model

Hazard and closure evidence

Geological Survey of India — landslide inventory · bhukosh.gsi.gov.in

ASDMA — flood and damage reports · asdma.gov.in

IMD — district rainfall warnings · mausam.imd.gov.in

NHAI / BRO / state PWD — closure and reopening bulletins, held as evidence

MODELS, STANDARDS AND METHODS

Fuel and emissions

CMEM — Comprehensive Modal Emissions Model (Barth et al., UC Riverside)

SAE J1321 Type II — fuel consumption test procedure

IPCC / MoEFCC diesel factor — 2.68 kg CO₂ per litre

Routing and geospatial

Luxen & Vetter, “Real-time routing with OpenStreetMap data”, ACM SIGSPATIAL 2011

PostGIS ST_DWithin — corridor buffering and stationary-vehicle detection

Security

Argon2id (RFC 9106) · OWASP refresh-token rotation with reuse detection

PostgreSQL Row Level Security — enabled on every table

OUR OWN RESEARCH FINDINGS

- ▶ Road-source survey: no Indian source publishes road reopening. That finding is why our own fleet traversals are treated as the strongest evidence a segment is open.
- ▶ Evidence governance: every route decision carries its inputs, its reason codes and the factors it could not obtain — recorded per segment, append-only.
- ▶ Field verification: installable Android APK exercised end-to-end against hosted PostGIS in ap-south-1; 1,636 automated tests across backend, manager and driver suites.

Every engine named is a deterministic weighted rule with published constants — auditable in the code, in the API response and on screen. Rules decide; the language model only restates what the rules produced.